

GOESGEN, Switzerland

WMO: 066260

Lat: 47.3631N

Lon: 7.9736E

Elev: 1247

StdP: 14.05

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.3	(c) 21.5	(d) 10.4	(e) 9.8	(f) 19.5	(g) 15.9	(h) 12.9	(i) 24.1	(j) 26.4	(k) 44.9	(l) 23.7	(m) 45.7	(n) 5.1	(o) 50	(p) 0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	88.9	68.4	85.1	67.5	81.5	66.0	70.3	84.1	68.8	81.5	67.3	78.6	5.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	101.1	74.8	64.6	95.6	72.3	63.4	91.4	70.9	35.0	84.3	33.6	81.9	32.4	78.8	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.6	(b) 14.6	(c) 11.8	(e) 11.6	(f) 94.3	(g) 5.5	(h) 2.6	(i) 7.7	(j) 96.1	(k) 4.5	(l) 97.7	(m) 1.5	(n) 99.1	(o) -2.5	(p) 101.0		
(a) 18.6	(b) 14.6	(c) 11.8	(e) 10.7	(f) 74.0	(g) 5.6	(h) 1.5	(i) 6.7	(j) 75.1	(k) 3.4	(l) 75.9	(m) 0.2	(n) 76.7	(o) -3.9	(p) 77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	34.4	35.2	43.0	50.8	57.1	64.7	68.4	67.3	59.7	51.6	42.5	35.8
	DBStd	13.60	6.58	7.35	6.13	6.20	6.34	5.99	5.54	5.63	5.46	6.45	6.27	6.29
	HDD50	1941	484	414	226	66	12	0	0	0	2	62	234	441
	HDD65	5468	948	834	683	427	255	78	30	42	173	416	676	906
	CDD50	2291	1	1	8	89	231	440	570	536	295	111	9	0
	CDD65	341	0	0	0	0	10	69	135	112	15	0	0	0
	CDH74	3294	0	0	0	23	153	726	1295	961	134	2	0	0
	CDH80	1081	0	0	0	1	23	235	476	326	20	0	0	0
Wind	WSAvg	3.8	4.9	4.3	4.7	3.7	3.8	3.3	3.2	2.7	2.9	3.0	3.9	4.7
Precipitation	PrecAvg	43.0	2.9	2.7	2.8	3.2	4.2	4.4	4.6	4.3	3.5	3.4	3.5	3.7
	PrecMax	54.4	7.4	6.6	9.8	7.0	7.2	9.1	8.0	8.4	6.2	6.4	8.3	7.8
	PrecMin	29.6	0.6	0.9	0.9	0.4	1.3	1.0	1.5	0.7	1.7	0.4	0.2	0.0
	PrecStd	6.4	1.7	1.5	1.9	1.8	1.7	1.9	1.5	1.8	1.4	1.6	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	59.8	67.2	77.7	83.4	91.9	94.5	92.1	82.8	72.0	62.4	52.9
		MCWB	47.9	48.0	51.8	57.5	65.3	70.9	69.7	69.3	65.6	59.6	53.8	48.5
	2%	DB	49.3	53.0	63.6	72.6	77.9	86.5	89.6	87.5	78.4	67.9	57.6	50.5
		MCWB	45.5	45.0	50.0	55.3	61.6	68.1	67.4	69.0	64.8	58.3	51.9	46.7
	5%	DB	46.1	48.7	59.5	69.3	74.6	81.7	85.4	83.7	74.0	64.0	54.3	47.6
		MCWB	42.6	43.4	48.3	54.6	59.8	66.2	67.1	67.9	62.3	57.1	49.9	44.4
	10%	DB	43.5	44.8	54.8	64.9	70.3	77.8	81.3	79.3	70.3	60.9	51.0	44.9
		MCWB	40.8	40.7	46.2	52.9	58.0	64.5	65.5	66.4	60.9	55.8	47.8	42.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	49.3	53.2	59.1	66.3	73.2	72.1	72.0	67.5	61.4	54.8	49.2
		MCDB	52.4	56.5	64.0	73.5	79.8	88.5	88.1	85.1	78.0	68.6	60.2	51.9
	2%	WB	45.9	45.7	50.9	57.1	63.7	69.8	69.9	70.1	65.5	59.5	52.5	47.1
		MCDB	48.7	52.4	61.1	69.3	74.7	83.1	84.3	83.3	75.3	65.5	57.1	50.0
	5%	WB	43.1	43.4	48.9	55.2	61.1	67.4	68.2	68.8	63.8	58.0	50.2	44.9
		MCDB	45.8	48.3	57.5	66.9	70.8	79.0	81.2	80.9	71.8	62.6	53.9	47.5
	10%	WB	41.1	40.9	47.0	53.4	59.2	65.5	66.8	67.2	62.0	56.4	48.2	42.4
		MCDB	43.4	44.5	54.1	63.9	68.3	75.7	78.3	77.6	68.4	60.1	50.4	44.5
Mean Daily Temperature Range	5% DB	MDBR	8.5	12.4	18.0	20.7	18.8	19.6	20.7	19.8	17.9	14.3	10.0	8.6
		MCDBR	12.7	20.9	28.3	30.6	27.4	27.9	29.3	27.1	24.2	19.4	15.4	11.6
		MCWBR	9.3	13.5	15.9	15.4	12.6	11.8	10.4	10.6	11.4	10.8	10.2	8.7
	5% WB	MCDBR	12.4	19.7	25.0	28.1	23.5	25.2	25.5	24.5	20.8	17.6	14.4	10.8
		MCWBR	9.4	12.9	14.5	14.5	11.3	11.2	9.9	10.3	10.3	10.3	9.8	8.5
Clear-Sky Solar Irradiance	taub	0.300	0.326	0.362	0.395	0.407	0.411	0.412	0.404	0.381	0.358	0.323	0.297	
	taud	2.440	2.367	2.316	2.257	2.268	2.300	2.291	2.327	2.374	2.421	2.464	2.448	
	Ebn at Noon	250	265	271	272	272	271	269	266	262	250	238	236	
	Edh at Noon	21	28	34	40	41	40	40	37	32	26	21	19	
All-Sky Solar Radiation	RadAvg	366	632	1017	1378	1588	1854	1786	1555	1177	742	406	299	
	RadStd	44	93	148	203	157	172	161	152	116	75	50	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (117 neighbors)	N/A	N/A	N/A	+1.31	+0.76	+0.58	N/A	N/A	+101	+25

Nomenclature: See separate page